



UNIVERSITY OF NAIROBI
SCHOOL OF COMPUTING AND INFORMATICS
SECOND SEMESTER EXAMINATIONS 2020/2021
SECOND YEAR EXAMINATIONS FOR THE BACHELOR OF SCIENCE IN COMPUTER
SCIENCE
CSC216: ASSEMBLY LANGUAGE PROGRAMMING
Date: Wednesday, 5th May 2021 Time: 9.00 AM -11.30 AM
Instruction: Answer Question 1 and any other two Questions

Registration No:

Question 1 (20 Marks)

A) Indicate how each of the following is achieved in an NAMS assembly language program [5 marks]

- i. Creation of a comment
- ii. Return an integer in a function
- iii. Return a float in a function
- iv. Return a string in a function
- v. Increment the value in eax by 1

B) Discuss the possible subdivisions of the EBX register [4 marks]

C) Give the instructions used to assemble, link and run a program in nasm. Assume the program file is hello.asm. [3 marks]

D) After some instruction, the parity flag is set. What does this imply? [2 marks]

E) As you write a program, you decide to set the direction flag

- i. What instruction will you use? [2 marks]
- ii. What will be the effect of this instruction? [2 marks]

F) Given the following instructions

MOV EBX, 10

What kind of addressing is used? [2 marks]

Question 2 (15 Marks)

Write an assembly language program with the following functionality

i. Prompts the user to enter two digits (each less than five)

ii. Gets the sum of the two digits

iii. Prints out the sum on the screen

Note: use a macros for all the outputs and inputs

Question 3 (15 Marks)

Write an assembly language program with the following functionality

i. Contains an array of 9 digits

ii. The program prints out the contents of the array in reverse (from the last to the first)

Question 4 (15 Marks)

Write an assembly language program with the following functionality

i. Adds two real numbers that are already initialised in the program

ii. Prints out the result

Note: You can use c functions (printf) within your assembly program

Question 5 (15 Marks)

Write an assembly language program with the following functionality

i. A function to get the square of a number passed to it and return the result

ii. The program calls the function defined in i using a number already defined in the program

iii. The program then prints the result returned in ii above.

Note: You can use c function (printf) within your assembly program